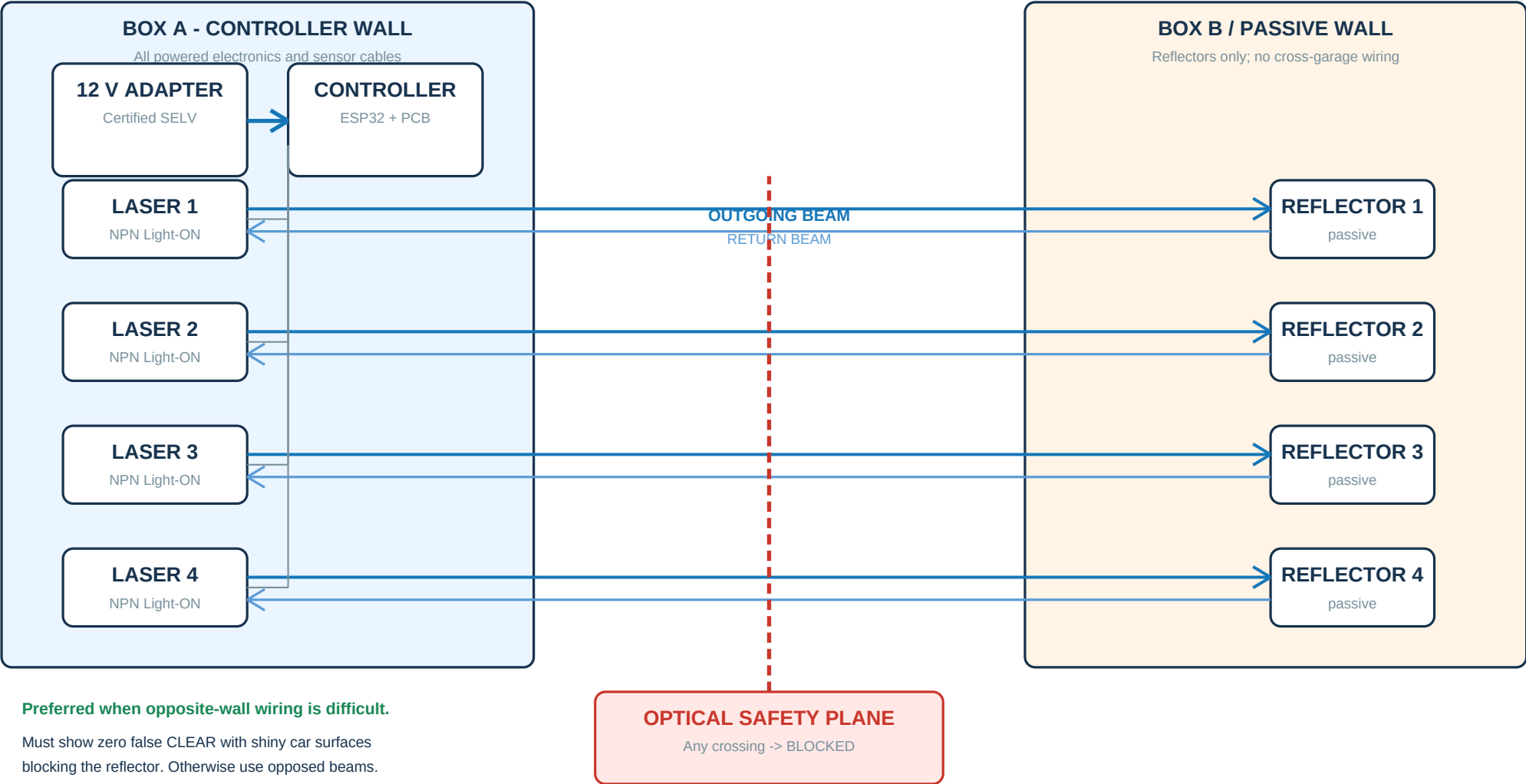
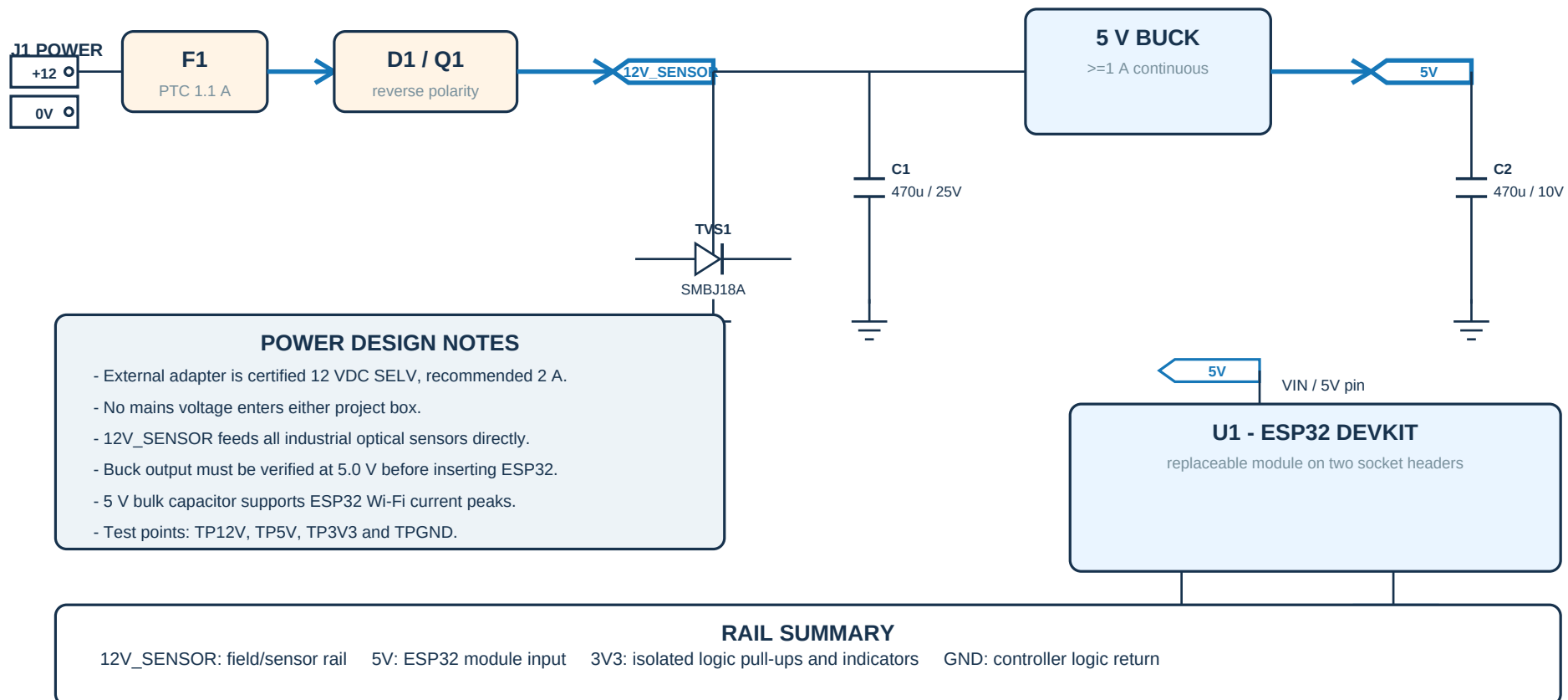


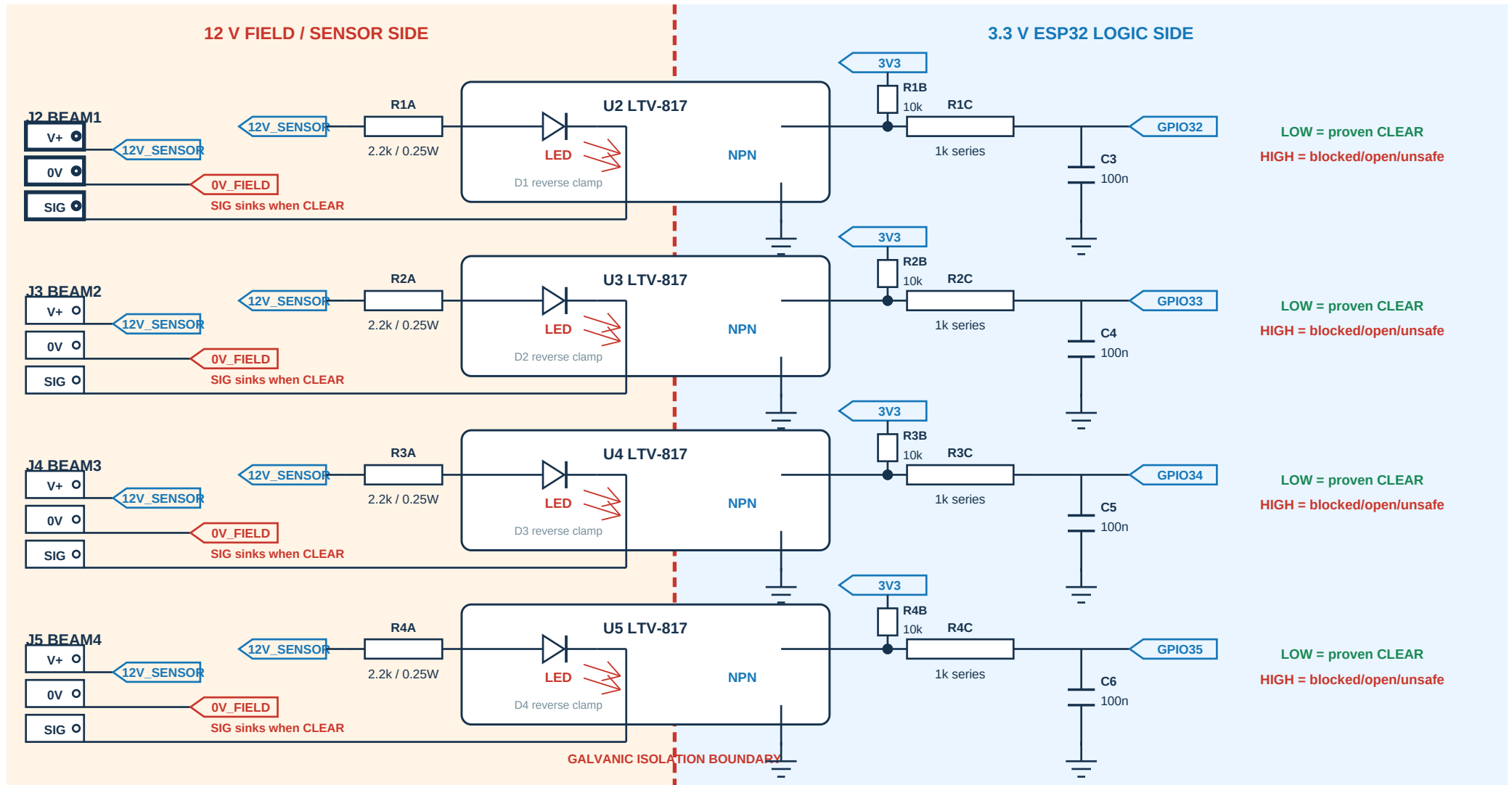
System architecture - same-side retroreflective option



Power input, protected rails and ESP32 module

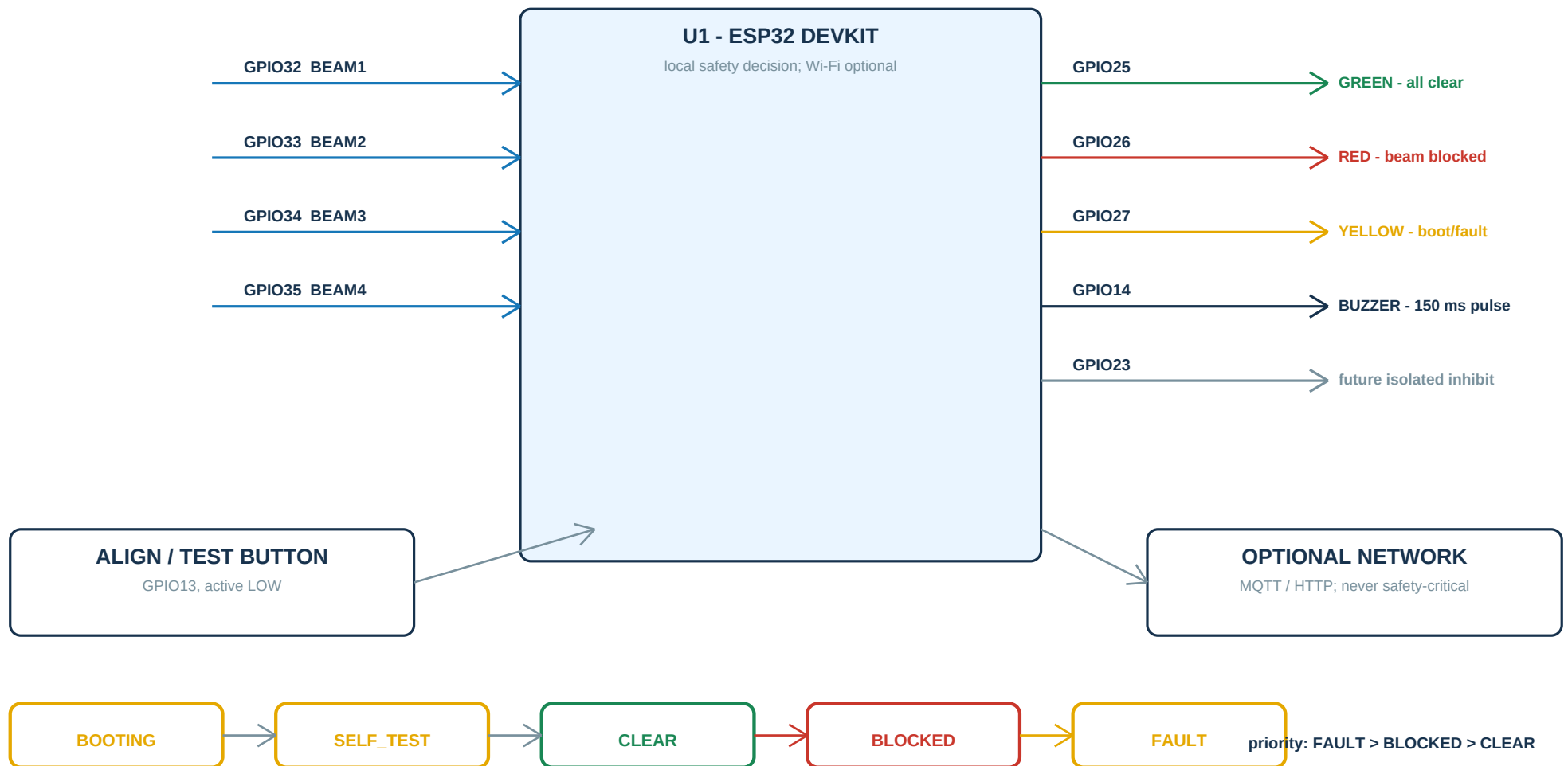


Four isolated fail-safe sensor inputs



D0 behavior: wire break or lost sensor power cannot earn CLEAR. A SIG-to-0V short requires the reserved power proof-test to detect.

ESP32 connections, indication and safety state behavior



CLEAR requires every enabled beam continuously clear for 500 ms. BLOCKED follows the 5 ms electrical noise filter. Startup never shows green.